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A General Energy-Balance Solver for Precooled Joule–Thomson Cryocoolers: Reproduction of Zhang & Dang (2021), Validation and Design Use

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Project report

Abstract

A solver is developed for the ideal steady-state analysis of Joule–Thomson (JT) cryocoolers precooled by a multi-stage cryocooler through a cascade of counterflow (CHEX) and precooling (PHEX) heat exchangers, for an arbitrary number of stages and any working fluid with tabulated or CoolProp properties. The model follows Zhang and Dang [Cryogenics 116 (2021) 103282]: it is an enthalpy-flow balance with prescribed exchanger effectiveness and homogeneous-equilibrium (HEM) choked flow through the orifice. We restate the model with all assumptions explicit, derive its main results (the cold-end refrigeration $q_C = \min(\Delta h_T, q_L)$, the optimum $\Delta h_T = q_L$, the effect of a finite effectiveness, and the global cascade balance), and describe the implementation, including He-3 property tables with monotone interpolation, an effectiveness definition valid on both sides of the optimum, an internal-pinch check and a direct HEM flux maximisation. Against the paper, with identical inputs, the solver gives $P_{U,\mathrm{opt}} = 0.358$ MPa (0.36), $q_{C,\mathrm{opt}} = 7.98$ kJ/kg (8.0), $\dot{m} = 2.339$ mg/s (2.34) and $Q_C = 18.62$ mW (18.7); the optimum fits agree to 1.4 % on average and the mass flux to better than 0.7 % on average. Applied unchanged to two other coolers of the same architecture, compared only with their ideal-model values, it reproduces Liu et al.'s calculated refrigeration to +1.1 % and brackets the design value of Ma et al. Finally the solver is used for design: since the upper precooler temperatures only redistribute a fixed total load, a load map and a Carnot-weighted criterion give the stage temperatures and loads a precooling cryocooler must provide.

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1 Introduction

A Joule–Thomson (JT) cooler produces refrigeration by throttling a high-pressure gas through a restriction; the temperature fall, and above all the refrigeration it can absorb at the cold end, depend strongly on the temperature the gas has before the restriction. Below about 4 K in He-4 and about 3 K in He-3 the gas has to be precooled to a few kelvin before throttling gives any useful effect. A *hybrid* (coupled) JT cooler does this with a cascade: the gas passes through a chain of counterflow heat exchangers (CHEX) in which the returning low-pressure vapour cools it, and between successive CHEXs a precooling heat exchanger (PHEX) is thermally anchored to a stage of a mechanical cryocooler (here a multi-stage pulse-tube cooler, SPTC). Zhang & Dang analyse such a cooler for He-3 at 1 K and give design rules.

Objectives. (a) Reproduce the paper's quantitative results from its own assumptions; (b) make the solver general enough to analyse other published coolers of the same architecture without modification; (c) use it to size the precooling stages of a new design. The report is organised as follows: system and notation (Section 2); model, assumptions and analytical results (Section 3); modelling decisions (Section 4); property models and implementation (Sections 5–6); validation against Zhang & Dang (Section 7) and against other coolers (Section 8); design (Section 9); limitations and reproducibility.

2 System and notation

2.1 Architecture

Figure 1 shows the architecture used by the paper (Fig. 1 there) and by every case in this document, for N CHEXs and $N-1$ PHEXs; the paper has $N=4$. High-pressure gas at P_U enters the warm end of CHEX 1 at T_H (300 K). It leaves CHEX i, is cooled by PHEX i to $T_{\mathrm{pre},i}$, and enters CHEX $i+1$. After the last CHEX it is throttled through the orifice (isenthalpic) to the low pressure P_D, partly evaporates at T_C in the evaporator (where the refrigeration Q_C is absorbed), and the saturated vapour returns through all CHEXs, cooling the incoming stream, to leave at the warm end.

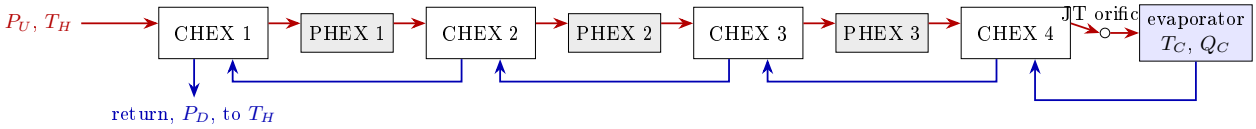


Figure 1: Architecture: red = high-pressure supply stream, blue = low-pressure return stream (schematic; cf. Fig. 1 of Zhang & Dang). PHEX j removes heat $Q_{\mathrm{pre},j}$ from the supply stream and is anchored to a stage of the precooling cryocooler.

2.2 State points

We follow the paper's numbering (its Fig. 2) for $N=4$. High-pressure side: 1 (CHEX 1 hot inlet, T_H), 2 (CHEX 1 hot outlet = PHEX 1 inlet), 3 (PHEX 1 outlet = CHEX 2 inlet, $T_{\mathrm{pre},1}$), 4, 5 ($T_{\mathrm{pre},2}$), 6, 7 ($T_{\mathrm{pre},3}$), 8 (CHEX 4 hot outlet, before the orifice). 9 is after the orifice (two-phase, $h_9 = h_8$), 10 is saturated vapour leaving the evaporator, 11, 12, 13 are the return-stream states between CHEXs and 14 is the warm-end exit. Points 1D, 3D, 5D, 7D are the states on the low-pressure isobar at the temperatures of points 1, 3, 5, 7. The code uses exactly this labelling for $N=4$ and generalises it for other N (Appendix A); Fig. 2 shows the resulting cycle in the P–h plane.

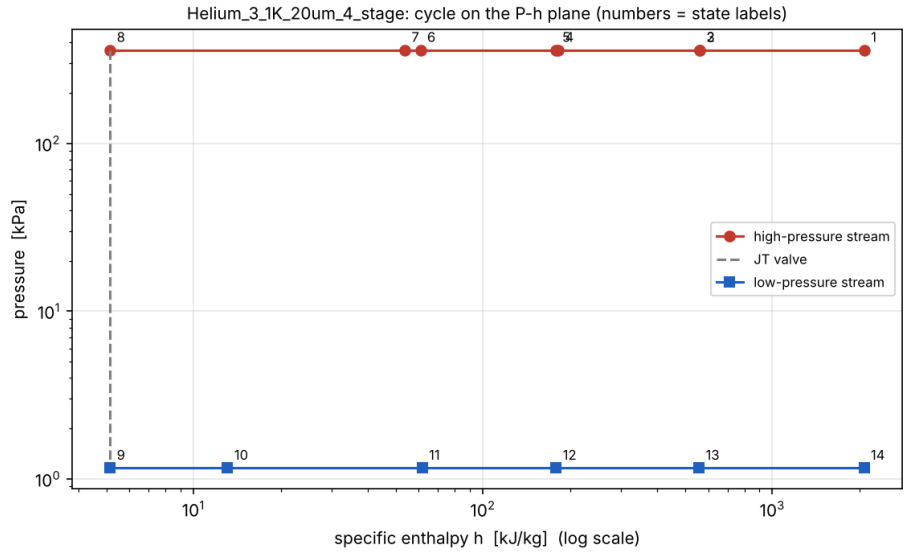


Figure 2: Cycle of the design point of the paper ($T_C = 1$ K, $P_U = 0.358$ MPa, $T_{\text{pre}} = (25.6, 12.3, 8)$ K, $d = 20$ µm) on the P–h plane (log–log), with the state labels. Output of `jt_run.py run`; see Section 9.

2.3 Symbols

$h,\ s,\ \rho$	specific enthalpy, entropy, density
$P_U,\ P_D$	high pressure; low pressure $= P_{\text{sat}}(T_C)$
$T_C,\ T_H$	evaporator (cold-end) temperature; warm-end temperature (300 K)
$T_{\text{pre},j}$	gas temperature leaving PHEX j (at P_U)
η_i	effectiveness of CHEX i (Section 3.3)
$q,\ Q$	per unit mass flow [kJ/kg], and power $Q = \dot{m}q$ [mW]
$q_C,\ Q_C$	refrigeration at the evaporator
$q_{\text{pre},j},\ Q_{\text{pre},j}$	heat removed at PHEX j
Δh_T	isothermal JT enthalpy difference $h(P_D, T_{\text{pre,last}}) - h(P_U, T_{\text{pre,last}})$
q_L	latent limit $h_g(T_C) - h(P_U, T_C)$
$G,\ d$	orifice mass flux [kg m^{-2}s^{-1}]; orifice diameter

3 Theory: the physical model and its assumptions

3.1 Assumptions, stated once

All results below follow from the following assumptions. Items marked $*$ are the paper's; the others are choices we made explicit because they matter when comparing with other work.

A1* Steady state; one-dimensional counterflow; the two streams carry the same mass flow $\dot{m}$ (no leaks between them, no mass storage).

A2* No heat in-leak from the surroundings and no axial conduction along the exchangers. All heat entering the system enters at the evaporator (Q_C) or leaves at the PHEXs.

A3* No pressure drop along either stream: the high-pressure stream is at P_U everywhere, the low-pressure stream at P_D everywhere.

A4* The orifice is adiabatic: $h_9 = h_8$.

A5* The evaporator is at saturation: the stream leaving it is saturated vapour at T_C, and $P_D = P_{\text{sat}}(T_C)$ (zero quality-independent temperature gradient in the evaporator).

A6* The precooler delivers the gas to a specified temperature $T_{\text{pre},j}$; how the cryocooler achieves this (its own capacity curve) is outside the model.

A7 Each CHEX is characterised by one number, the effectiveness η_i, defined from an enthalpy balance, not by UA and length. Properties vary through the exchanger as the real fluid dictates (no ideal-gas shortcuts).

A8 The mass flow is set by the orifice (choked flow of the real fluid), or given from a measurement. It is never obtained from a compressor model.

A9 No discharge-coefficient correction ($C_d = 1$) and no multi-orifice effects.

Assumptions A2, A3 and A9 are optimistic and are why a real cooler delivers less than the model (Sections 8 and 10).

3.2 Energy balance of one counterflow exchanger

Label the four ends of a CHEX: hot stream in A (at P_U), hot stream out B; cold stream in C (at P_D), cold stream out D. With A1–A3 the first law for each stream gives the same heat duty Q per unit mass flow:

$$h_B = h_A - Q, \qquad h_D = h_C + Q. \tag{1}$$

Two physical limits bound Q. The *warm-end limit*: the cold stream cannot leave warmer than the hot stream enters, so $h_D \le h(P_D, T_A)$, i.e.

$$Q \le Q_{\text{warm}} = h(P_D, T_A) - h_C. \tag{2}$$

The *cold-end limit*: the hot stream cannot leave colder than the cold stream enters, $h_B \ge h(P_U, T_C)$ with T_C now the temperature of state C:

$$Q \le Q_{\text{cold}} = h_A - h(P_U, T_C). \tag{3}$$

$Q_{\max} = \min(Q_{\text{warm}}, Q_{\text{cold}})$. This is the enthalpy (not temperature) form of the effectiveness definition and it is essential for real gases: because the specific heat of the high-pressure stream differs strongly from that of the low-pressure stream (Fig. 3), the two limits are generally different and *which one binds is a result, not an assumption.* For He-3 at 0.36 MPa, c_p rises to about 12 kJ/kg/K near 5 K and falls to about 1.3 kJ/kg/K at 1 K, while at 1.16 kPa $c_p \approx 7$ kJ/kg/K throughout; the high-pressure stream has the larger capacity rate above ≈ 4 K and the smaller one below it, which is why the last CHEX is cold-end limited and the others are warm-end limited (this is the "pinch" discussion of the paper, its Fig. 4).

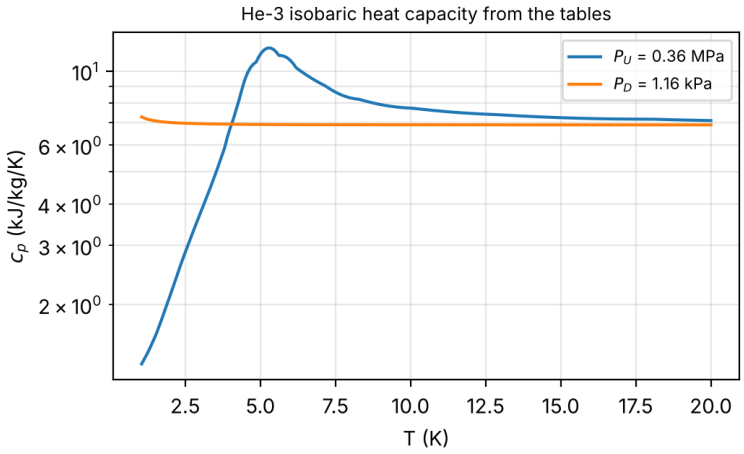


Figure 3: He-3 isobaric heat capacity from the tables at $P_U = 0.36$ MPa and $P_D = 1.16$ kPa (cf. Fig. 4 of the paper). Computed from `h(P,T)` by finite differences.

3.3 Effectiveness: the paper's definition and ours

The paper defines the effectiveness of the last CHEX from the cold side only (its Eq. 4):

$$\eta_4 = \frac{h_{11} - h_{10}}{h_{7D} - h_{10}}. \tag{4}$$

This is Q/Q_{warm}. We define instead

$$\eta = \frac{Q}{Q_{\max}}, \qquad Q_{\max} = \min(Q_{\text{warm}}, Q_{\text{cold}}). \tag{5}$$

The two coincide whenever the exchanger is warm-end limited, and for the paper's last CHEX at and below the optimal P_U (the regime of its analysis). When P_U is above the optimum the last CHEX is cold-end limited, $Q_{\text{cold}} < Q_{\text{warm}}$, and Eq. (4) with $\eta < 1$ would permit $Q > Q_{\text{cold}}$, i.e. $T_B < T_C$: the hot stream leaves colder than the cold stream enters, which is unphysical. Definition (5) is therefore used throughout,

and we state this as a deliberate, checkable difference from the paper (it does not change any number in the paper's regime; see the validation in Section 7). The same definition makes η_i directly comparable with the effectiveness quoted by other authors from measured temperatures (Section 3.9).

3.4 The last stage: $q_C = \min(\Delta h_T,\, q_L)$

Consider the last CHEX, orifice and evaporator (states 7–11, paper Fig. 3). The refrigeration absorbed per unit mass is the enthalpy rise in the evaporator,

$$q_C = h_{10} - h_9 = h_{10} - h_8, \tag{6}$$

with $h_{10} = h_g(T_C)$ and $h_8 = h_7 - Q$ from Eq. (1). For an ideal CHEX ($\eta = 1$) $Q = Q_{\max}$ and Eq. (6) becomes

$$\text{warm-end limited: } q_C = h_{10} - h_7 + (h_{7D} - h_{10}) = h(P_D, T_7) - h(P_U, T_7) \equiv \Delta h_T, \tag{7}$$

$$\text{cold-end limited: } q_C = h_{10} - h(P_U, T_C) \equiv q_L. \tag{8}$$

Because the limiting case is the one with the smaller Q, which is the one giving the larger h_8 and hence the *smaller* q_C,

$$q_C = \min(\Delta h_T,\, q_L), \tag{9}$$

which is the paper's Eqs. (30)–(31). Δh_T is the isothermal JT enthalpy difference at the precooling temperature (how much enthalpy the gas gives up when throttled isothermally at T_7); q_L is the latent heat of the liquid at the evaporator temperature compressed to P_U (the most the liquid could ever absorb on evaporation).

Dependence on pressure. From $(\partial h/\partial P)_T = v(1 - \beta T)$ (β the volume expansivity),

$$\Delta h_T(P_U) = \int_{P_D}^{P_U} v\,(\beta T - 1)\,\mathrm{d}P, \qquad q_L(P_U) = h_g - h(P_U, T_C), \quad \frac{\mathrm{d}q_L}{\mathrm{d}P_U} = -v_L(1 - \beta_L T_C) < 0. \tag{10}$$

For the liquid $\beta_L T_C \ll 1$, so q_L falls almost linearly with P_U and vanishes at a maximum pressure $P_{U,\max}$ (1.131 MPa for He-3 at 1 K; paper: 1.13 MPa), above which no refrigeration is possible at any precooling temperature. Δh_T rises with P_U (below the inversion temperature) and saturates. Hence $q_C(P_U)$ increases with Δh_T until the two curves cross, then decreases along q_L: the maximum is at the crossing,

$$\Delta h_T(P_{U,\mathrm{opt}}, T_{\mathrm{pre}}) = q_L(P_{U,\mathrm{opt}}), \tag{11}$$

the paper's Eq. (34), and equality of the two is both necessary and sufficient for the optimum (Fig. 4, left). Equation (11) is one equation in two unknowns, so it defines a curve $P_{U,\mathrm{opt}}(T_{\mathrm{pre}})$ (paper's Fig. 8, fitted by its Eqs. 35–36).

Dependence on T_{pre}. Δh_T has a maximum in temperature (near 4 K for He-3 at 0.36 MPa, dashed green curve in Fig. 4, right) and falls strongly as T_{pre} rises; q_L does not depend on T_{pre}. While $\Delta h_T > q_L$ the refrigeration is on the plateau $q_C = q_L$: cooling the gas further does not help. The *warmest* T_{pre} on the plateau is therefore the best choice, $T_{\mathrm{pre,opt}}$: it delivers the same q_C with the least demanding precooler. At $P_U = 0.358$ MPa this is 8 K, which is the paper's choice of $T_{\mathrm{pre3}} = 8$ K.

Imperfect last CHEX. For $\eta < 1$, $Q = \eta Q_{\max}$ and Eq. (6) gives

$$q_C = \eta\, Q_{\max} - (h_7 - h_{10}) = \eta\, \min(\Delta h_T, q_L) - (1 - \eta)\,(h_7 - h_{10}). \tag{12}$$

The penalty is $(1 - \eta)$ times the *whole* enthalpy difference between the warm hot-stream end and the cold return stream (here $h_7 - h_{10} = 40.7$ kJ/kg), about five times the useful q_C. Hence a 1 % loss of effectiveness costs about 6 % of q_C (we find 7.96 → 7.47 kJ/kg for $\eta = 1 \to 0.99$), which is why the last CHEX is the critical component. Together with the associated mass-flow change (Section 3.6), this gives the large sensitivity of Q_C shown in Section 7.

3.5 Cascade balance and what the upstream precoolers can change

For ideal exchangers the paper obtains, from Eq. (1) applied to each CHEX and the precoolers (its Eqs. 18–22),

$$q_C = \sum_j q_{\mathrm{pre},j} + \big[h(P_D, T_H) - h(P_U, T_H)\big], \tag{13}$$

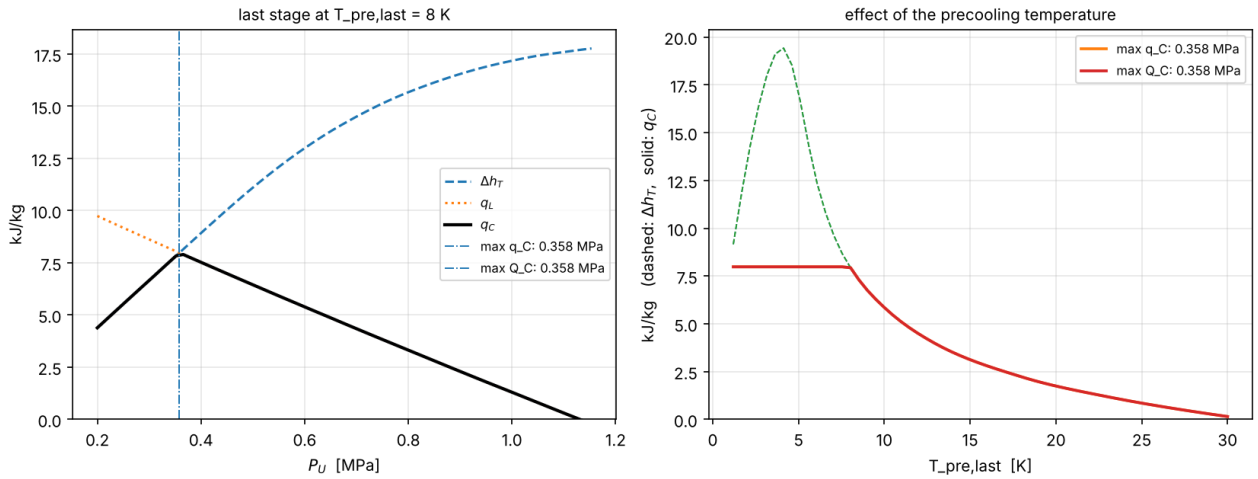


Figure 4: Last stage of the design example. Left: Δh_T, q_L and $q_C = \min$ versus P_U at $T_{\mathrm{pre,last}} = 8$ K; the optimum is the crossing at 0.358 MPa and the curve reaches zero at 1.13 MPa (cf. paper Fig. 5). Right: effect of $T_{\mathrm{pre,last}}$ (at the optimal P_U the plateau $q_C = 7.98$ kJ/kg ends at 8 K; cf. paper Fig. 7).

where the bracket is the (negative) isothermal JT enthalpy difference at ambient, and $Q_C = \dot{m} q_C$. Equation (13) is just the first law over the whole machine: everything the precoolers remove, plus the net enthalpy the compressed gas brings in, balances the refrigeration. For $\eta < 1$ the same balance holds with $h(P_D, T_{D1})$ in place of $h(P_D, T_H)$, T_{D1} being the actual warm-end return temperature; the solver evaluates it that way and checks it.

An important consequence. The last CHEX sees fixed boundary states on both streams: its hot inlet $(T_{\mathrm{pre,last}}, P_U)$ and its cold inlet (saturated vapour at T_C). Its duty, q_C, T_8, and through T_8 the orifice flux and $\dot{m}$, are therefore determined by T_C, P_U, $T_{\mathrm{pre,last}}$, η_{last} and the orifice alone. *The upstream precooler temperatures cannot change q_C, $\dot{m}$ or Q_C*; they only decide how the (fixed) total precooling load $\sum_j q_{\mathrm{pre},j}$ in Eq. (13) is distributed among the PHEXs. This separates the design problem into two stages: optimise the last stage (Section 3.7), then choose the upstream temperatures by a different criterion (Section 3.8). We verified the invariance numerically (Fig. 12: the total load and Q_C stay constant while any one upstream temperature is swept over its full range).

3.6 Mass flow: choked flow through the orifice

$Q_C = \dot{m} q_C$ needs $\dot{m} = \pi d^2 G/4$, with G the mass flux through the orifice. The paper computes G for real He-3 from the choked-flow condition using the fundamental derivative-based factor Γ and the real-gas equation of state (its Eqs. 41–48), and fits the results (Table 2) with an algebraic form (Eq. 49) as function of P_U and T_8. We use an equivalent but simpler formulation that needs only the property tables: a *homogeneous-equilibrium model* (HEM). For stagnation state (P_0, h_0, s_0) (state 8) and an isentropic expansion to pressure P,

$$G(P) = \rho(P, s_0)\sqrt{2\left[h_0 - h(P, s_0)\right]}, \qquad G_{\mathrm{choked}} = \max_{P_D \le P < P_U} G(P). \tag{14}$$

The maximum is found on 500 logarithmically spaced pressures and refined by golden-section search. If the maximum lies at $P = P_D$ the flow is not choked (flag `UNCHOKED_RESTRICTION_FLOW`); this is typical when the inlet is liquid-like. Equation (14) agrees with the paper's real-gas results to 0.25–0.72 % (mean) at $T_C = 1, 2, 2.6, 3$ K (Table 1). The paper's own fit (Eq. 49) is also implemented (`--flow-model paper_fit`) and is itself reproduced by our HEM data to 0.2–1.3 % in the worst case.

Why $\dot{m}$ depends on η. G depends on the stagnation state 8; with $\eta < 1$, T_8 rises (from 1 K to 1.91 K at $\eta = 0.97$), the fluid is less dense and its flux falls, hence the decrease of $\dot{m}$ with η in the paper's Fig. 15 (2.339 → 2.267 mg/s here).

3.7 Optimising Q_C rather than q_C

Since $Q_C = \dot{m} q_C$ and $\dot{m}$ increases with P_U (flux grows with upstream pressure), the pressure for maximum Q_C is above that for maximum q_C *when the flux gain outweighs the q_C loss*. In the paper's case with $T_{\mathrm{pre}} = 5$ K

the q_C maximum is at 0.21 MPa but the Q_C maximum is at 0.36 MPa (paper, Sect. 5); we obtain 0.213 and 0.352 MPa (Section 7). A second subtlety is that q_C alone has *no* interior optimum in $T_{\mathrm{pre,last}}$ (it is monotone: colder is never worse), so "the optimum" only exists with a constraint or a different objective. We therefore (i) optimise Q_C when a flow is available, with a "practical" option that picks the warmest T_{pre} and the lowest P_U whose Q_C is within 0.5 % of the maximum (a flat optimum: Fig. 5), and (ii) use $T_{\mathrm{pre,opt}}$ of Eq. (11) when T_{pre} is to be chosen.

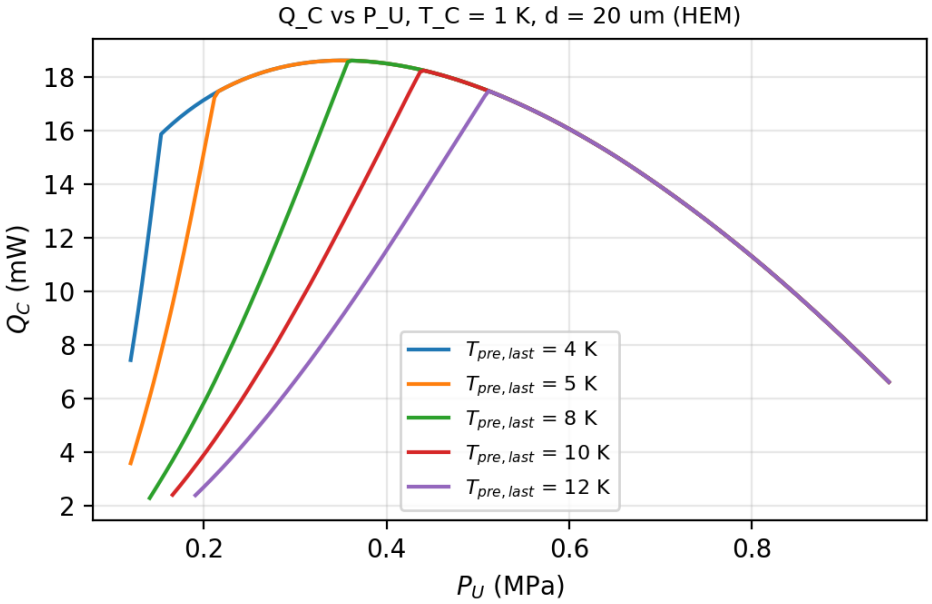


Figure 5: Q_C versus P_U at $T_C = 1$ K, $d = 20$ µm (HEM flux) for several $T_{\mathrm{pre,last}}$ (cf. Fig. 13 of the paper). Curves with colder T_{pre} reach the common envelope (maximum 18.6 mW at about 0.36 MPa) at lower P_U; warmer T_{pre} forces the optimum to higher P_U and lower Q_C.

3.8 Choosing the upstream temperatures

By Section 3.5 upstream temperatures $T_{\mathrm{pre},j}$ ($j < N-1$) leave q_C unchanged, so another criterion is needed. Physically, a precooler at temperature T_j rejecting Q_j to ambient T_H needs at least the Carnot work $Q_j(T_H/T_j - 1)$, and real cryocoolers are much more demanding at low T_j. We therefore minimise

$$\Phi = \sum_j Q_{\mathrm{pre},j}\left(\frac{T_H}{T_j} - 1\right) \tag{15}$$

over the upstream $T_{\mathrm{pre},j}$, subject to $T_H > T_1 > T_2 > \cdots > T_{\mathrm{last}}$ and every $Q_{\mathrm{pre},j} \geq 0$ (a PHEX may not heat the gas). A differential-evolution search is used, because Φ has non-smooth regions where a PHEX load reaches zero. This is only one defensible criterion (the real choice depends on the capacity curves of the cryocooler used, Section 9), so the code also produces the full map of loads versus each upstream temperature, from which the user can choose any other.

3.9 Prescribed flow and effectiveness from temperatures

Some papers give $\dot{m}$ and temperatures rather than an orifice diameter. The solver accepts a prescribed `mdot_mg_s` instead of d (d and $\dot{m}$ together are rejected as over-specified). If the paper reports the temperatures at the two ends of the last CHEX (e.g. hot stream in at T_A, return stream out at T_D), the effectiveness follows from Eq. (5) in the warm-limited case,

$$\eta = \frac{h(P_D, T_D) - h_C}{h(P_D, T_A) - h_C}. \tag{16}$$

This is used for the Liu et al. case (Section 8).

3.10 Internal pinch check

Eq. (1)–(5) constrain only the ends of an exchanger. With strongly varying c_p the stream temperatures could still cross *inside*. The solver therefore integrates the exchanger at 20 equal enthalpy steps, computing both stream temperatures, and flags a negative gap (`INTERNAL_TEMPERATURE_CROSSOVER`); this is a warning, not an error. In the paper's design CHEX 1 shows a small crossover (minimum gap -0.01 K): a sign that the ideal $\eta = 1$ is physically marginal in that exchanger, not a solver failure.

4 Modelling decisions

The following decisions determine what the solver does and does not claim.

1. **Energy balance over UA models.** The paper works in the P–h plane with effectiveness; a UA–NTU model would need geometry and heat-transfer correlations that the target papers do not provide. All inputs are therefore thermodynamic: $T_C, P_U, T_{\mathrm{pre},j}, \eta_i, d$ or $\dot{m}$.
2. **Properties from the paper's own tables.** For He-3 the supplied single-phase and saturation tables (not an equation of state) are the single source of truth; this removes the ambiguity of reference states and keeps the reproduction faithful. For other fluids CoolProp is used through the same interface.
3. **Everything generic in N.** The cycle is solved for any number of CHEXs, not only four, so other coolers (3 CHEX with 2 PHEXs, etc.) run unchanged.
4. **HEM for the orifice** (reasoned above), with the paper's own fit available for comparison.
5. **Fail loudly.** Inputs outside the tables raise an error (no extrapolation); the feasibility check explains infeasibility rather than returning a number.
6. **Validation first, claims second.** The solver was checked against all quantitative statements of the paper before it was applied to other papers; other-paper cases carry their reported values and a reference block, and mismatches are explained rather than tuned away (no free fitting parameter is used to force agreement).
7. **One front door.** A single command-line script (no GUI) drives validation, other papers and own designs; every run is stored with its configuration so that a result can always be traced to inputs.

5 Property models

5.1 He-3 from the supplied tables

The paper's results depend on He-3 properties in a region (0.5–3.3 K, 1 Pa–3 MPa) where no standard equation of state package is available, so the solver reads two CSV tables (`data/he3/`): single-phase properties on a (T, P) grid (0.5–300 K, 1 Pa–3 MPa) and saturation properties (0.5–3.2 K). The loader checks columns, units, monotonicity and splits the saturation table into liquid and vapour branches; `python jt_run.py check` reports the SHA-256 of both files so a run can be tied to a data version (here `b17901e3e299` and `8a674a6a03da`) and confirms $P_{\mathrm{sat}}(1\,\mathrm{K}) = 1160.3$ Pa, which is the paper's 1.16 kPa.

5.2 Interpolation

- **Method.** $h(P,T)$, $s(P,T)$, $\rho(P,T)$ are interpolated with monotone piecewise-cubic Hermite polynomials (PCHIP), first along each tabulated isotherm in $\ln P$, then across isotherms in T at the requested pressure. PCHIP does not overshoot: crucial for $c_p = \partial h/\partial T$ where c_p has a sharp peak (Fig. 3).
- **Phase classes.** The table spans the liquid and vapour phases. For each (P,T) the phase is decided by comparing T with $T_{\mathrm{sat}}(P)$; interpolation is done only among nodes of the same class so that the discontinuity at saturation is not smoothed over. Above the critical point (3.3246 K, 116.4 kPa) a single class is used; the critical point is used for *classification*, closing the saturation dome.
- **Near-critical gap.** Between 101.7 kPa (last tabulated saturation point) and the critical pressure the tables have no saturation data; requests there raise `OutOfRange`. This is why the pressure sweeps of the Q_C study start at 0.12 MPa. A liquid-class lookup needs at least three nodes, hence it is not available below ≈ 45 Pa.
- **No extrapolation.** Anything outside the tables raises `OutOfRange`; T_C is limited to ≤ 3.2 K. This is deliberate (Section 4, item 5).

- **Inverse.** $T(P,h)$ is obtained by bracketed root finding on $h(P,T)$; a $T \to h \to T$ round trip at 0.36 MPa, 8 K returns 8.00000 K (part of the data check). Absolute values of h and s depend on the table's reference state; only differences enter any result, so the reference state is not checked.

5.3 Other fluids

For He-4, Ne, H_2, para-H_2, N_2, Ar, O_2, CH_4 and others, the same interface (`h, s, rho, Psat, T_from_Ph, ...`) is served by CoolProp. Two restrictions follow from the tables-free approach: the high pressure must exceed the critical pressure of the fluid (He-4 0.227 MPa is not an issue, but Ne 2.66, H_2 1.30, N_2 3.40 MPa are), otherwise the hot stream condenses inside the exchangers and the property lookup fails; and the feasibility check restricts T_C from the triple point to $0.995\,T_{\mathrm{crit}}$.

6 Implementation

6.1 Structure

File	Content
`jt/properties.py`	`Helium3Fluid`: CSV-backed tables, PCHIP interpolation, saturation, inverse, error classes
`jt/cycle_core.py`	state points, the CHEX solver (Eqs. 1–5), `solve_cycle`, the HEM choked flux (Eq. 14), the CoolProp adapter, flags
`jt/design_optimize.py`	last-stage optimisation of P_U, $T_{\mathrm{pre,last}}$; Eq. 49 fit; case-file naming
`jt/phex_load_map.py`	PHEX loads versus an upstream temperature
`jt/workflow.py`	configuration normalisation, feasibility check, optimiser pipeline, upstream search, run storage
`jt_run.py`	the command-line front door
`run_all.py`	one-command validation chain (data check, paper, other papers)
`reproduce_zhang_dang_2021.py`	reproduction of the paper's figures/fits
`tests/test_workflow.py`	29 unit tests

6.2 Solving a cycle (`solve_cycle`)

Given T_C, P_U, T_H, $T_{\mathrm{pre},1..N-1}$, $\eta_{1..N}$ and d or $\dot{m}$:

1. Compute $P_D = P_{\mathrm{sat}}(T_C)$ and the evaporator outlet $h_{10} = h_g(T_C)$.
2. **Last CHEX first.** Its hot inlet $(T_{\mathrm{pre,last}}, P_U)$ and cold inlet (h_{10}) are known; apply Eqs. (1)–(5) to get Q, hence h_8, h_{11} and the states on both streams. $q_C = h_{10} - h_8$.
3. **Work upstream.** The cold-stream inlet of CHEX i is the outlet of CHEX $i+1$ (known). The hot inlet of CHEX i is $(T_{\mathrm{pre},i-1}, P_U)$ (or T_H for $i = 1$). Apply Eqs. (1)–(5). The PHEX load follows from the enthalpies on either side: $q_{\mathrm{pre},i} = h(\text{CHEX } i \text{ hot out}) - h(P_U, T_{\mathrm{pre},i})$.
4. $\dot{m} = \pi d^2 G/4$ from Eq. (14) at the stagnation state 8 (or the given $\dot{m}$); $Q_C = \dot{m} q_C$, $Q_{\mathrm{pre},j} = \dot{m} q_{\mathrm{pre},j}$.
5. **Checks.** global balance Eq. (13), non-negative PHEX loads, temperature ordering, internal pinch scan, choking.

Because every step is explicit (no iteration on the cycle), a solve takes milliseconds, which makes the scans of Section 9 cheap; the cost is in the property lookups and the choked-flux search.

6.3 Theory-to-code map

Equation / statement	Paper	Where in the solver
CHEX energy balance, limits (1)–(3)	Eqs. 2, 5, 9–12	CHEX solver in `cycle_core.py`
Effectiveness (5) (differs from Eq. 4 beyond optimum)	Eq. 4	CHEX solver, `eta`
$q_C = \min(\Delta h_T, q_L)$ (9)	Eqs. 30–31	`solve_cycle` (emerges, not coded); tested by the scorecard check "$q_C = \min(\Delta h_T, q_L)$"
Optimum $\Delta h_T = q_L$ (11)	Eq. 34	`design_optimize.py` (scan, then root refinement)
Fits $P_{U,\mathrm{opt}}$, $q_{C,\mathrm{opt}}$	Eqs. 35–36, Table 1	compared in `reproduce_zhang_dang_2021.py`
Cascade balance (13)	Eq. 22	global-balance check in `solve_cycle`
HEM flux (14)	Eqs. 41–48	`cycle_core.py`
Paper's flux fit	Eq. 49, Table 2	`design_optimize.py`, `--flow-model paper_fit`
$Q_C = \dot{m} q_C$	Eq. 50	`solve_cycle`
Upstream criterion (15)	— (new)	`workflow.optimise_upstream`
Upstream sensitivity map	— (new)	`workflow.upstream_map`
Feasibility	— (new)	`workflow.check_config`

6.4 Feasibility check

`jt_run.py feasible case.json` answers "is this design physically possible and why not". In order: (1) T_C inside the fluid window (He-3 0.5–3.2 K; CoolProp fluids triple point to $0.995 T_{\mathrm{crit}}$); (2) $P_U > P_D$; (3) T_{pre} strictly decreasing from T_H and all above T_C; (4) $q_L = h_g(T_C) - h(P_U, T_C) > 0$, otherwise no refrigeration at any T_{pre} (the message gives $P_{U,\max}$); (5) $\Delta h_T > 0$ at $T_{\mathrm{pre,last}}$, otherwise the gas is above its inversion temperature; (6) a warning if $T_{\mathrm{pre,last}} > T_{\mathrm{pre,opt}}$ (then $q_C = \Delta h_T$ is limited by the precooling); (7) a trial solve whose flags are classified as errors (a PHEX would have to heat the gas, balance fails, no net refrigeration) or warnings (internal pinch, unchoked flow). The verdict is *feasible*, *feasible with warnings* or *infeasible*.

6.5 Run storage, naming and the data reference

- **Case names.** `<Fluid>_<T_C>K_<d>um_<N>_stage[_eta<x>]`, e.g. `Helium_3_1K_20um_4_stage`; a prescribed flow replaces the orifice by `<mdot>mgs`, `nod` marks "no flow information", and `_v2`, `_v3` are appended if the name already exists with different content. He-4 is labelled `Helium_4`.
- **Run folders.** Every `run` writes `results/runs/<config name>/<stamp>_<solve|optimize>/` with `config.json`, `meta.json`, `result.json`, `states.csv`, `ph_cycle.png` and, for optimisation, `map_qC.png`, `map_QC.png`, `stage_curves.png`, `upstream_map.png/csv`.
- **Linking runs to configurations.** `meta.json` holds a hash of the input configuration (fluid, T_C, P_U, T_H, T_{pre}, η, d or $\dot{m}$, optimiser settings). The data-reference block (citation, DOI, location in the paper, kind, note) is *excluded* from the hash, so a reference can be added or corrected later without orphaning the runs. `python jt_run.py runs` lists the runs against configurations.

6.6 Command reference

Command	Purpose
`check`	He-3 tables present and consistent
`validate [-quick]`	workflow 1: data check, tests, paper reproduction, scorecard
`new -fluid ...`	create a named case with data reference, check feasibility
`feasible cases/x.json`	feasibility verdict and reasons
`run cases/x.json [-optimize ...]`	solve, or optimise if anything is blank
`runs`	list saved runs and their configurations
`solve cases/*.json`	plain solve with reported-value comparison and sweeps
`optimize, phex`	original last-stage optimiser and PHEX load map
`python run_all.py [-quick]`	everything in one command

Blank inputs (- for T_{pre} or P_U) trigger the optimiser; the options are `--drive Q_C|Q_C_practical|q_C`, `--flow-model hem|paper_fit`, `--pu-min/--pu-max`, `--tpre-min/--tpre-max`, `--hem-grid`, `--save-final`.

6.7 Where the generated files go

Location (relative to the project folder)	Content
`results/validation/`	paper reproduction report, scorecard (`scorecard.json`), figures
`results/validation_all/`	`run_all.py` summary (`summary.csv`, `summary.json`), per-case results
`results/cases/<name>/`	output of `solve` (result, states, sweeps)
`results/runs/<config>/<stamp>_<kind>/`	output of `run` (Section above)
`cases/`	case files (inputs, references, reported values)
`paper_figs/ (delivered with this document)`	the figures and data used here

`results/` is generated output and can be deleted at any time.

7 Validation against Zhang & Dang (2021)

All values are produced by `python jt_run.py validate --dense` (curves with 241 points, optimisation grids of 61 points, 40 precooling temperatures, 46 flux points, effectiveness step 0.0025). Inputs are those of the paper: He-3, $T_C = 1$ K, $T_H = 300$ K, $d = 20$ µm, $T_{\mathrm{pre}} = (80, 25, 8)$ K. In each comparison figure, panel (a) is extracted directly from the paper and panel (b) is the solver output for the same inputs; the paper's figures are not redrawn.

7.1 Last-stage optimum (paper Figs. 5–8)

Quantity	Solver	Paper	Difference
$P_{U,\mathrm{opt}}$ at $T_{\mathrm{pre3}} = 8$ K [MPa]	0.3584	0.36	−0.4 %
$q_{C,\mathrm{opt}}$ [kJ/kg]	7.979	8.0	−0.3 %
$P_{U,\mathrm{max}}$ ($q_L = 0$) [MPa]	1.131	1.13	+0.1 %
P_U for max q_C, $T_{\mathrm{pre}} = 5$ K [MPa]	0.213	0.21	+1.4 %
P_U for max Q_C, $T_{\mathrm{pre}} = 5$ K [MPa]	0.352	0.36	−2.2 %
$\Delta h_T = q_L$ at the optimum (Eq. 11), max. rel. gap	7.5×10^{-8}	—	identity
$q_C = \min(\Delta h_T, q_L)$, max. abs. deviation	4×10^{-14}	—	identity

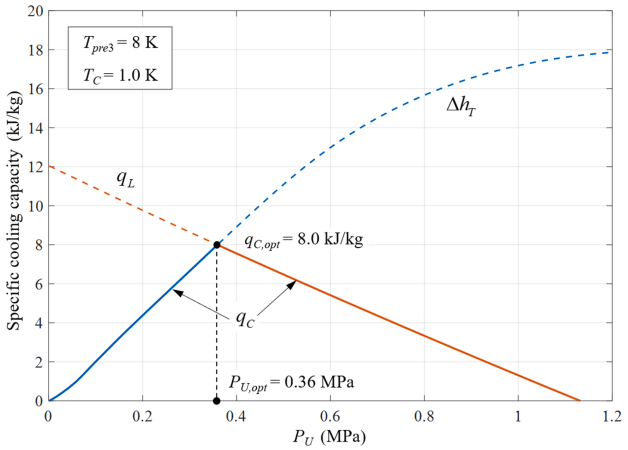
The last two rows confirm that the minimum rule and the optimum condition emerge from the energy balance; neither is imposed in the code.

Correlations of the optimum (paper Eqs. 35–36). Over the paper's range of T_{pre3} the solver's $P_{U,\mathrm{opt}}$ differs from Eq. (36) by 1.39 % on average and 3.72 % at worst, and $q_{C,\mathrm{opt}}$ from Eqs. (35)–(36) by 3.12 % on average (largest deviation 10.4 % at 30 K, where q_C is small). Evaluated at the solver's own $P_{U,\mathrm{opt}}$, the paper's q_C correlation agrees to 0.17 % on average (0.47 % worst). For the optimum precooling temperature versus P_U the differences from Eq. (36) at 0.3, 0.5, 0.7 and 0.9 MPa are 0.01, 0.23, 0.34 and 0.78 K. No optimum lies on the boundary of the search range.

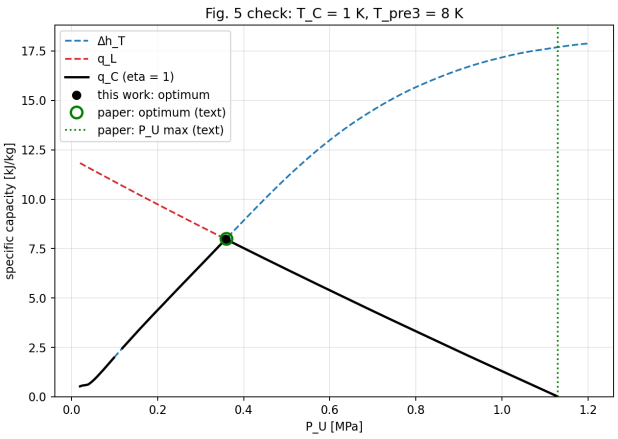
7.2 Mass flux (paper Eqs. 41–49, Table 2, Fig. 11)

7.3 Gross cooling capacity (paper Fig. 13)

Combining q_C with the HEM flux through $d = 20$ µm gives $Q_C = \dot{m}\, q_C$. With $T_{\mathrm{pre3}} = 8$ K the maximum of Q_C coincides with that of q_C ($P_U = 0.358$ MPa, $Q_C = 18.62$ mW); for a colder last precooler the Q_C maximum moves to a higher pressure than the q_C maximum because $\dot{m}$ increases with P_U (at 5 K: 0.352 vs. 0.213 MPa).

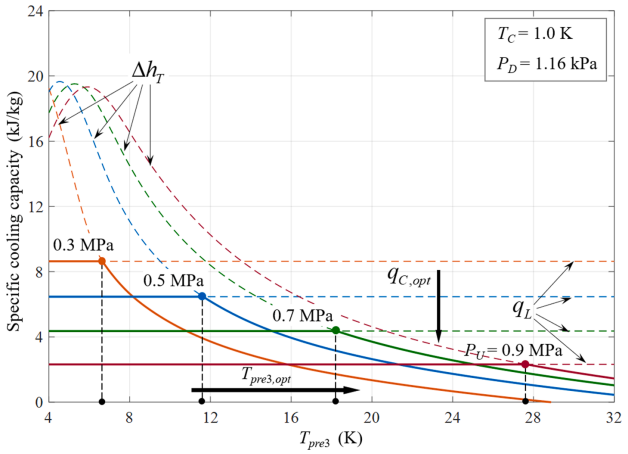


(a) Zhang & Dang (2021)

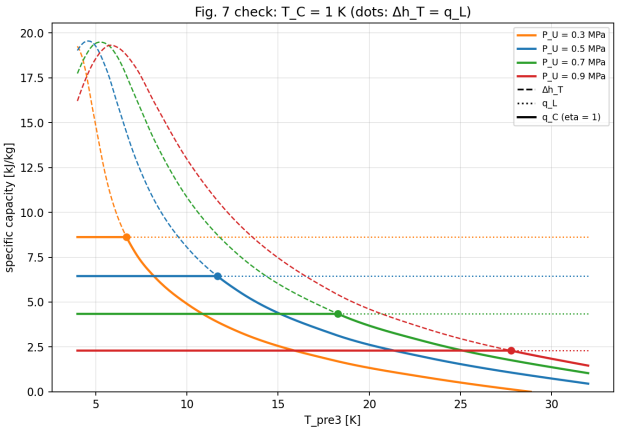


(b) This work, identical inputs

Figure 6: Refrigeration per unit mass q_C versus P_U at $T_{\mathrm{pre3}} = 8$ K (paper Fig. 5). The open circle marks the optimum.



(a) Zhang & Dang (2021)



(b) This work, identical inputs

Figure 7: q_C versus T_{pre3} for several P_U (paper Fig. 7).

7.4 Effect of the CHEX effectiveness (paper Figs. 14–16)

Keeping all other CHEXs ideal and varying the last effectiveness (optimal $P_U = 0.358$ MPa, $T_{\mathrm{pre3}} = 8$ K, $d = 20$ µm):

η_4	T_8 [K]	T_{11} shortfall [K]	q_C [kJ/kg]	$\dot{m}$ [mg/s]	Q_C [mW]
1	1.000	0	7.961	2.339	18.62
0.995	1.180	0.043	7.718	2.332	18.00
0.99	1.349	0.078	7.475	2.322	17.36
0.98	1.651	0.149	6.989	2.297	16.06
0.97	1.910	0.219	6.503	2.267	14.74
0.95	2.333	0.360	5.531	2.197	12.15
0.90	3.086	0.712	3.102	1.994	6.19

("T_{11} shortfall" is $T_7 - T_{11}$, the temperature by which the return stream leaves the CHEX colder than the warm-end temperature.) Comparison with the paper: $Q_C(\eta = 1) = 18.62$ vs. 18.7 mW, $\dot{m} = 2.339$ vs. 2.34 mg/s, $q_C = 7.96$ vs. 8.0 kJ/kg; $Q_C(\eta = 0.97) = 14.74$ vs. 14.7 mW and a relative drop of 20.8 % vs. about 21 % (paper Fig. 16). The paper's statement that for $\eta_4 \geq 0.97$ the temperature differences at the cold and warm ends should not exceed 1 K and 0.2 K (its Fig. 14) is met: we find 0.91 K ($T_8 - T_C$) and 0.22 K ($T_7 - T_{11}$) at $\eta = 0.97$. The sensitivities of Q_C are about -7 % per 0.01 of η near $\eta = 1$ and the change is almost linear (paper Fig. 16); the contribution of the flux ($\dot{m}$ drops by 3 %) is small compared with that of q_C (18 %) consistent with Eq. (12).

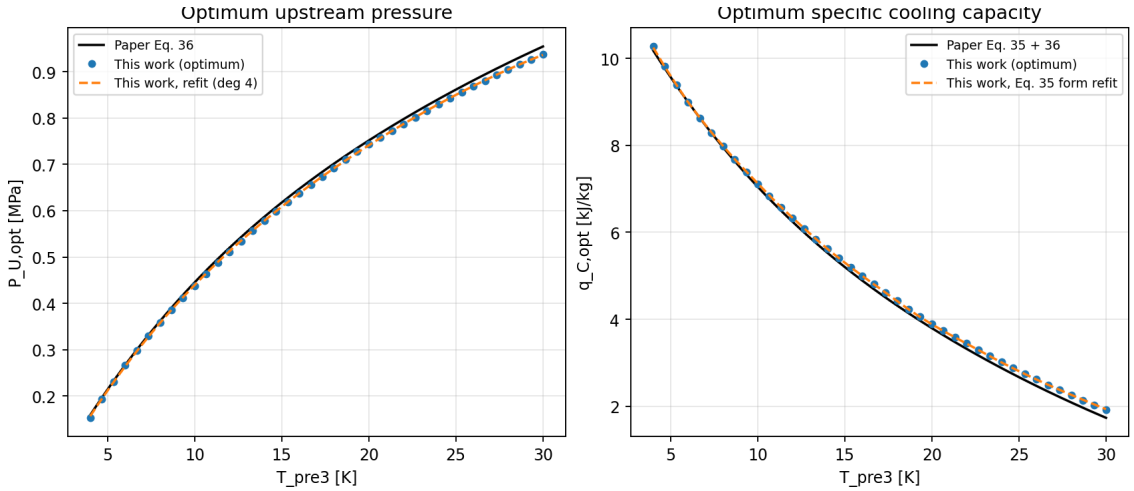


Figure 8: Solver optima (markers) against the paper's correlations Eqs. (35)–(36) (lines): $P_{U,\mathrm{opt}}$ and $q_{C,\mathrm{opt}}$ versus T_{pre3}.

T_C [K]	1	2	2.6	3
HEM flux vs. paper Table 2, mean deviation [%]	0.25	0.41	0.51	0.69
HEM flux vs. paper Table 2, worst deviation [%]	0.71	0.72	2.28	1.49
Eq. 49 form refitted to HEM data, worst deviation [%]	0.24	0.53	1.00	1.82

Table 1: HEM mass flux against the paper's real-gas flux.

7.5 Precooling loads for the paper's temperatures

At $P_U = 0.36$ MPa and $T_{\mathrm{pre}} = (80, 25, 8)$ K the solver gives $Q_{\mathrm{pre}} = (0.511, 9.698, 16.613)$ mW and $Q_C = 18.62$ mW, $\dot{m} = 2.339$ mg/s; i.e. the 8 K stage has to absorb about 16.6 mW. Cascade balance Eq. (13) is satisfied to rounding. The solver flags an internal temperature crossover in CHEX 1 (min. gap -0.013 K) and the unchoked flow (the inlet is liquid-like at 8 K, so the HEM maximum is at the back pressure; the flux is still the HEM maximum).

7.6 Summary

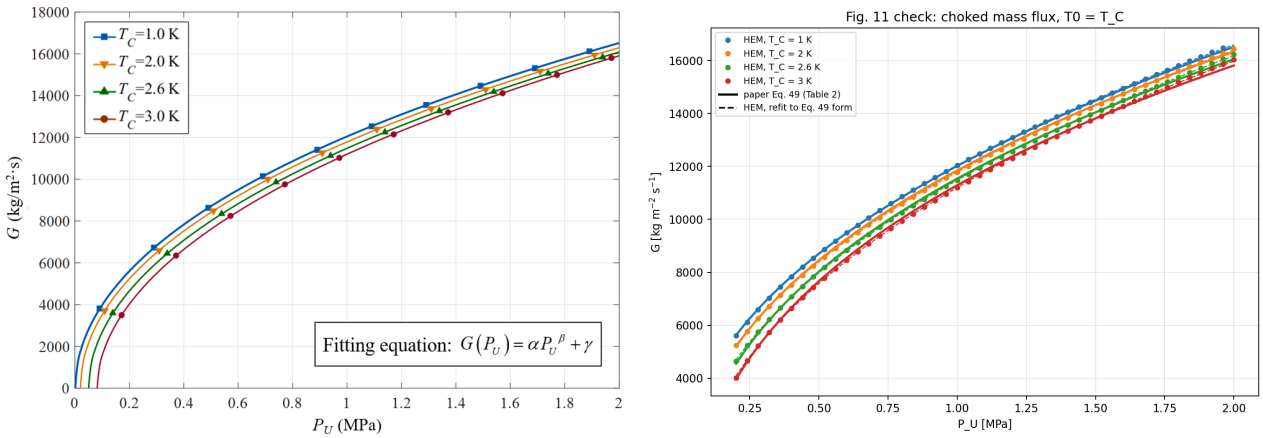
The validation comprises 27 checks of three kinds: *anchors* (values read from the paper), *consistency* checks (identities the implementation must satisfy) and *model* checks (HEM against the paper's real-gas flux). All pass; the complete list is given in Appendix D. The largest deviation of a directly reported value is 2.2 %.

8 Application to other coolers of the same architecture

To test generality the solver, unchanged, was applied to coolers from two further papers. Only *ideal-model* values reported by those authors are compared, i.e. values computed under the same assumptions (no heat leak, no pressure drop, prescribed effectiveness). Measured performance is deliberately excluded: a measurement contains parasitic loads and line pressure drops that the model omits by construction, so a comparison with it tests the model's assumptions rather than its implementation. The working fluid in both cases is He-4 (CoolProp).

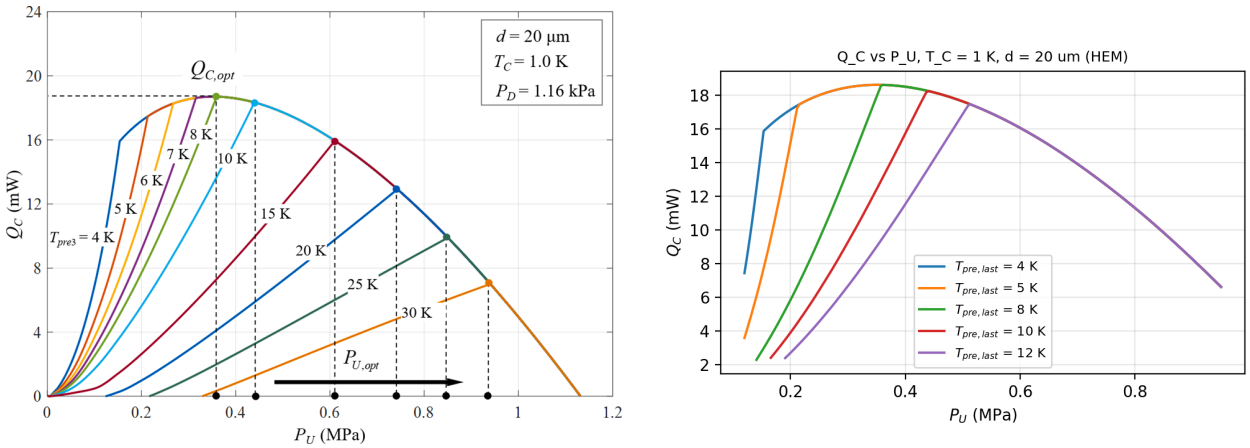
Case	T_C [K]	P_U [MPa]	$\dot{m}$ [mg/s]	T_{pre} [K]; η
Liu et al. (2017)	4.252	1.137	9.1	($60^{\dagger}$, 17.3); ($1^{\dagger}$, $1^{\dagger}$, 0.9857)
Ma et al. (2019), design point	4.5	2.0	10.5	(100, 15); 0.97–1

Table 2: Inputs († value not given by the source; it affects only the split of precooling load, not q_C, $\dot{m}$ or Q_C).



(a) Zhang & Dang (2021) (b) This work, identical inputs

Figure 9: Maximum mass flux versus P_U for several T_C (paper Fig. 11).



(a) Zhang & Dang (2021) (b) This work, identical inputs

Figure 10: Gross cooling capacity Q_C versus P_U for several last-stage precooling temperatures (paper Fig. 13).

Liu et al. (2017), Int. J. Heat Mass Transf. 109, doi:10.1016/j.ijheatmasstransfer.2017.02.070. A He-4 JT stage with supply 1.137 MPa, return 0.104 MPa ($T_C = T_{\mathrm{sat}} = 4.252$ K), $\dot{m} = 9.1$ mg/s and last-stage precooling at 17.3 K; the authors' calculated maximum refrigeration is 52.5 mW. No effectiveness is reported, but the supply enters the last exchanger at 17.3 K and the return leaves it at 17.1 K, so $\eta_3 = 0.9857$ follows from Eq. (16) without fitting. The solver gives $q_C = 5.835$ kJ/kg and $Q_C = 53.1$ mW (+1.1 %). With $\eta_3 = 1$ the result would be about 19 % higher, so the agreement depends on the effectiveness derived from the reported temperatures.

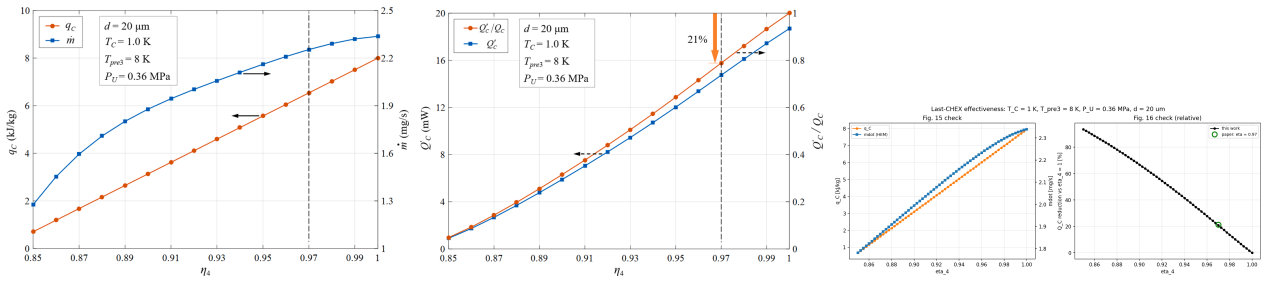
Ma et al. (2019), Cryogenics 104, doi:10.1016/j.cryogenics.2019.102983, design point: 100 mW at 4.5 K, 2.0 MPa, precooling at 100 and 15 K, effectiveness "not lower than 0.97". Over that admissible range the solver gives 90.2 mW ($\eta = 0.97$) to 108.8 mW ($\eta = 1$); the design value corresponds to $\eta \approx 0.985$. The load of the 15 K precooler, 278 mW, is within 7 % of the design value of 260 mW.

9 Application: design of a coupled hybrid JT cooler

The validated solver is used the way a designer would use it: choose the target, find the operating point, and read off what the precooling cryocooler has to supply.

9.1 Procedure

1. Create a named case: `python jt_run.py new --fluid he3 --T_C 1.0 --eta 1,1,1,1 --T_pre=-,-,8 --P_U=- --d_um 20`. This produces `Helium_3_1K_20um_4_stage.json`. (Here the last stage temperature is given, 8 K, and the rest is left to the optimiser.)



(a) Paper Fig. 15 (b) Paper Fig. 16 (c) This work

Figure 11: Effect of the last-CHEX effectiveness η_4 on q_C, $\dot{m}$ and Q_C.

Case	Solver	Reported (ideal)	Difference
Liu 2017, Q_C [mW]	53.1	52.5	+1.1 %
Ma 2019 design, Q_C [mW], $\eta = 0.97$–1	90.2–108.8	100	bracketed ($\eta \approx 0.985$)
Ma 2019 design, Q_{pre} at 15 K [mW]	278	260	+7 %

Table 3: Solver results against the ideal values reported by the source papers.

2. `python jt_run.py feasible cases/Helium_3_1K_20um_4_stage.json`: verdict, $P_{U,\max} = 1.131$ MPa, $T_{\text{pre,opt}}$.
3. `python jt_run.py run cases/Helium_3_1K_20um_4_stage.json`: optimise and solve.

9.2 Result for the paper's cooler

The optimiser finds $P_U = 0.3584$ MPa and $Q_C = 18.62$ mW (both for q_C and Q_C), with $T_{\text{pre,opt}} = 8$ K at that pressure, without being told the optimum; it then selects the upstream temperatures by criterion (15):

PHEX	Gas in [K]	Gas out (T_{pre}) [K]	q [kJ/kg]	Q [mW]
1	26.24	25.64	4.240	9.90
2	12.77	12.29	3.564	8.32
3	8.446	8.00	3.665	8.56
Total precooling			11.47	26.77

The refrigeration $q_C = 7.979$ kJ/kg, $\dot{m} = 2.334$ mg/s (the small difference from 2.339 is the optimal P_U 0.3584 vs. 0.36 MPa). The state table is printed by `run` and stored in `states.csv`; $T_8 = 1$ K, $T_{11} = 8$ K, $h_8 = 5.13$ kJ/kg.

9.3 The upstream map: how to select the precooling cryocooler

Because the upstream temperatures only redistribute a fixed total of 11.47 kJ/kg (26.8 mW), the useful design information is the *map* of loads against temperature. Figure 12 sweeps PHEX 1 (left) and PHEX 2 (right) with the others fixed. Reading it:

- Total precooling load is constant (dashed black).
- Raising $T_{\text{pre},1}$ moves load from PHEX 1 to PHEX 2; above ≈ 90 K PHEX 1 carries almost nothing (the exchanger 1 already brings the gas to that temperature) and PHEX 2, the second stage, carries about 18 mW.
- PHEX 2 at 8–25 K trades load with PHEX 3: a colder $T_{\text{pre},2}$ moves load from PHEX 3 to PHEX 2.

The Carnot-weighted cost Φ (Eq. 15) is 263 at the chosen $T_{\text{pre},1} = 26.7$ K (this is the minimum of the sweep; 316 at 12 K, 305–316 above 70 K) and 264 at $T_{\text{pre},2} = 11.1$ K. For a real cryocooler the capacity curve decides: choose the cooler whose stage capacities at its stage temperatures meet or exceed the loads in this map. For the paper's cooler a four-stage cryocooler with stages near 26, 12 and 8 K carrying about 10, 8 and 9 mW (plus the first stage at 80–300 K carrying no significant load) satisfies the optimum.

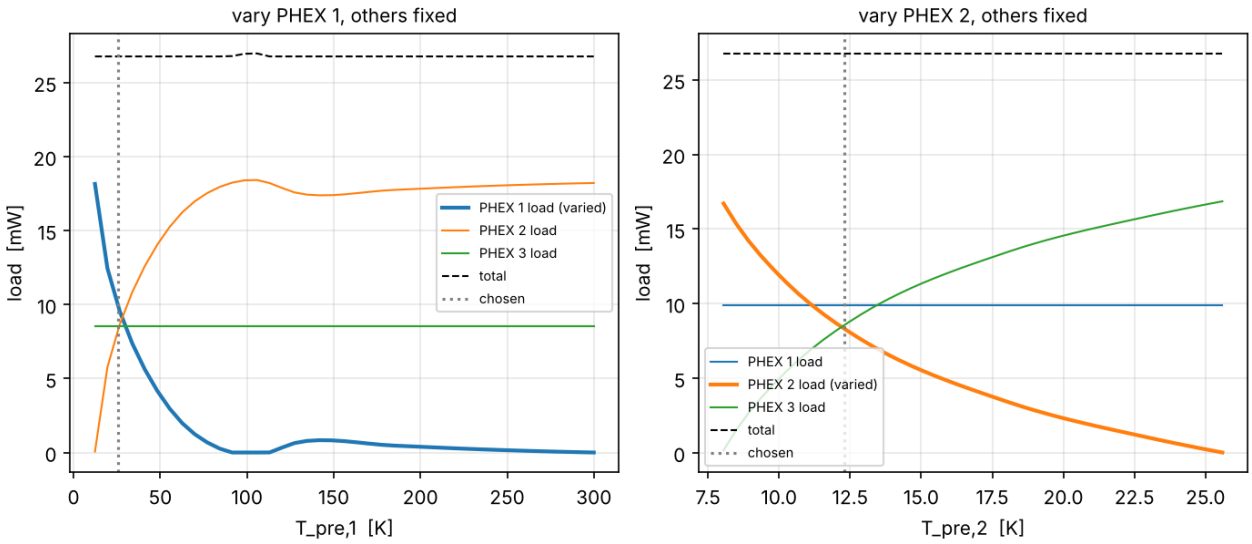


Figure 12: Loads of the three PHEXs when PHEX 1 (left) or PHEX 2 (right) temperature is varied; total constant; the dotted line marks the chosen value.

9.4 Comparison with the paper's precooling temperatures

With the paper's choice $T_{\text{pre}} = (80, 25, 8)$ K at $P_U = 0.36$ MPa the loads are (0.5, 9.7, 16.6) mW: the total is the same, but most of it falls on the coldest stage. The Carnot-weighted choice (25.6, 12.3, 8) K spreads it as (9.9, 8.3, 8.6) mW and reduces the work-equivalent cost, since load at 8 K is the most expensive to remove.

9.5 Using the same machinery for another fluid or target

The same three commands work for any supported fluid: e.g. He-4 at 4.5 K, or Ne. The feasibility check will report $P_{U,\max}$, the inversion limit and $T_{\text{pre,opt}}$, so one can see directly whether a target is possible and how warm the last-stage precooler can be, before any optimisation. The cases of Section 8 can be re-run with `--optimize` to obtain the best T_{pre} and P_U for those coolers.

10 Limitations

- **Ideal model.** No heat leak, no pressure drop, no axial conduction, discharge coefficient $C_d = 1$ (A2, A3, A9). Results are upper bounds for a real cooler, and the solver is therefore compared only with ideal-model values.
- **Effectiveness is an input.** It is not predicted from exchanger geometry; Q_C falls by about 7 % per 0.01 of η_{last} near $\eta = 1$, so η must come from measurement or a separate exchanger model.
- **Precooling is imposed.** Cryocooler capacity curves are not modelled; the load map states the capacities required.
- **Mass flow** is the HEM maximum through a single ideal orifice, or a prescribed value; compressor coupling and multi-orifice elements are not modelled.
- **He-3 property range:** $T_C \leq 3.2$ K, no extrapolation, a near-critical gap at 101.7–116.4 kPa.
- **Other fluids:** P_U must exceed the critical pressure; accuracy is that of the CoolProp equations of state.
- **Internal pinch** is detected and reported, not corrected.
- **Upstream criterion.** Carnot weighting is one reasonable choice; the actual cryocooler capacity curve should decide.

11 Reproducibility

1. `pip install numpy scipy pandas matplotlib CoolProp`; place the He-3 tables in `data/he3/`.
2. `python jt_run.py validate --dense`: paper reproduction, all figures and the 27-check scorecard (grid sizes can be set individually with `--n-curve`, `--n-tpre`, `--n-opt-grid`, `--n-flux`, `--eta-step`).

3. `python make_extra_figures.py --n-pu 200 --outdir results/extra`: Q_C versus P_U curves and He-3 c_p (with progress display).
4. `python run_all.py`: data check, paper reproduction and the other-cooler cases.
5. Design: `python jt_run.py new --fluid he3 --T_C 1.0 --eta 1,1,1,1 --T_pre=-,-,8 --P_U=- --d_um 20`, then `feasible` and `run` on the created case file.

A State labelling for general N

For N CHEXs the hot-stream states are numbered $1, \ldots, 2N$ in order along the supply path (CHEX i hot inlet $2i-1$, hot outlet $2i$; the following PHEX leaves state $2i+1$), orifice outlet $2N+1$, evaporator outlet $2N+2$, then the return-stream states $2N+3, \ldots, 4N+2$ from the cold end of CHEX N to the warm exit. For $N = 4$: 1–8 hot, 9 orifice outlet, 10 evaporator outlet, 11–14 return (the paper's numbering, its Fig. 2). The `states.csv` file holds T, P, h, phase and a role string for each.

B Flags

`Flag`	Meaning
`INTERNAL_TEMPERATURE_CROSSOVER`	stream temperatures cross inside a CHEX (warning)
`UNCHOKED_RESTRICTION_FLOW`	HEM maximum flux at the back pressure; flux still the HEM maximum (warning)
`(error class)`	a PHEX would have to heat the gas; balance fails; no net refrigeration

C Example case file

```
{ "name": "Liu 2017 He-4 open-cycle JT, last stage at 17.3 K",
  "fluid": "Helium", "T_C": 4.2517, "P_U": 1.137, "T_H": 300,
  "T_pre": [60.0, 17.3], "eta": [1.0, 1.0, 0.9857],
  "mdot_mg_s": 9.1,
  "reference": {"citation": "...", "location": "Sec. 2.2 / Eq. (1)",
                "kind": "...", "note": "..."},
  "reported": {"Q_C_mW": 52.5, "P_D_kPa": 104.0} }
```

Keys beginning with `_` are ignored; `d_um` and `mdot_mg_s` are mutually exclusive.

D Scorecard of the paper reproduction

Check	Solver	Paper	Tolerance
$P_{U,\mathrm{opt}}$ at $T_{\mathrm{pre3}} = 8$ K [MPa]	0.3584	0.36	± 0.03
$q_{C,\mathrm{opt}}$ [kJ/kg]	7.979	8.0	± 0.4
$P_{U,\max}$ [MPa]	1.131	1.13	± 0.05
$q_C = \min(\Delta h_T, q_L)$	4×10^{-14}	—	$\leq 10^{-6}$
$\Delta h_T = q_L$ at optimum (Eq. 34)	7.5×10^{-8}	—	≤ 0.01
$P_{U,\mathrm{opt}}$ vs. Eq. 36, mean / worst [%]	1.39 / 3.72	—	5 / 10
$q_{C,\mathrm{opt}}$ vs. Eqs. 35+36, mean [%]	3.12	—	5
$T_{\mathrm{pre3,opt}}$ at 0.3/0.5/0.7/0.9 MPa [K]	0.01/0.23/0.34/0.78	Eq. 36	± 1.5
HEM flux at $T_C = 1, 2, 2.6, 3$ K, mean [%]	0.25/0.41/0.51/0.69	Table 2	15
Eq. 49 form, worst [%]	0.24/0.53/1.00/1.82	—	2
q_C ($\eta = 1$) [kJ/kg]	7.961	8.0	± 0.4
Q_C ($\eta = 1$) [mW]	18.62	18.7	± 1
$\dot{m}$ ($\eta = 1$) [mg/s]	2.339	2.34	± 0.15
Q_C ($\eta = 0.97$) [mW]	14.74	14.7	± 1
Q_C drop at $\eta = 0.97$ [%]	20.82	21.39	± 3

The table groups related checks; there are 27 in total, all of which pass.

References

[1] T. Zhang, H. Dang, Investigations on a 1 K hybrid cryocooler composed of a four-stage Stirling-type pulse tube cryocooler and a Joule–Thomson cooler. Part A: Theoretical analyses and modeling, *Cryogenics* 116 (2021) 103282, doi:10.1016/j.cryogenics.2021.103282.

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[3] Ma et al., *Cryogenics* 104 (2019) 102983, doi:10.1016/j.cryogenics.2019.102983.

[4] I. H. Bell et al., Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp, *Ind. Eng. Chem. Res.* 53 (2014) 2498–2508.